URJA-CHAKRA

ऊर्जा चक्र

Unified **R**egistry for **J**oule **A**ccounting

HAR WATT KA HISAAB · ACCOUNT FOR EVERY WATT

A Government of India energy credit portal. Factories and commercial buildings stream live meter data, receive a machinery-aware savings target, and earn tradable energy credits for every unit of electricity they save.

1 UEC = 1,000 kWh of verified electricity saved

PROBLEM STATEMENT
Intelligent Energy Consumption Optimization System

PS ID / THEME
[PS ID] · Clean & Green Technology
Category: Software

TEAM
[Team Name] · Team ID [____]
JECRC Foundation, Jaipur

EVENT
Smart India Hackathon 2026
NITK Nodal Centre

Industry saves energy for free — and gets nothing back for it

41.8%

of India's total electricity consumption comes from the industrial sector — the single largest share of any sector.

IBEF / CEA, FY 2024

WHAT EXISTS

The **PAT scheme** rewards energy saving with tradable ESCerts — but only for **Designated Consumers**, the largest plants in 13 sectors.

WHO IS LEFT OUT

4.7 crore registered MSMEs, commercial buildings, malls and campuses have **no energy incentive mechanism at all**.

WHY IT IS FAILING

PAT verifies on a **three-year cycle** from audited paperwork, and its certificate market cleared only at the floor price.

THE CORE GAP

A factory that cuts 10,000 units of electricity this month gets nothing but a smaller bill. There is no national rail that measures that saving as it happens and pays for it.

ENERGY THAT LEAKS AWAY BETWEEN TWO AUDIT CYCLES

Idle & off-shift running

Motors, compressors and lines drawing load while producing nothing.

Compressed-air leaks

The costliest utility in a plant, leaking continuously and unnoticed.

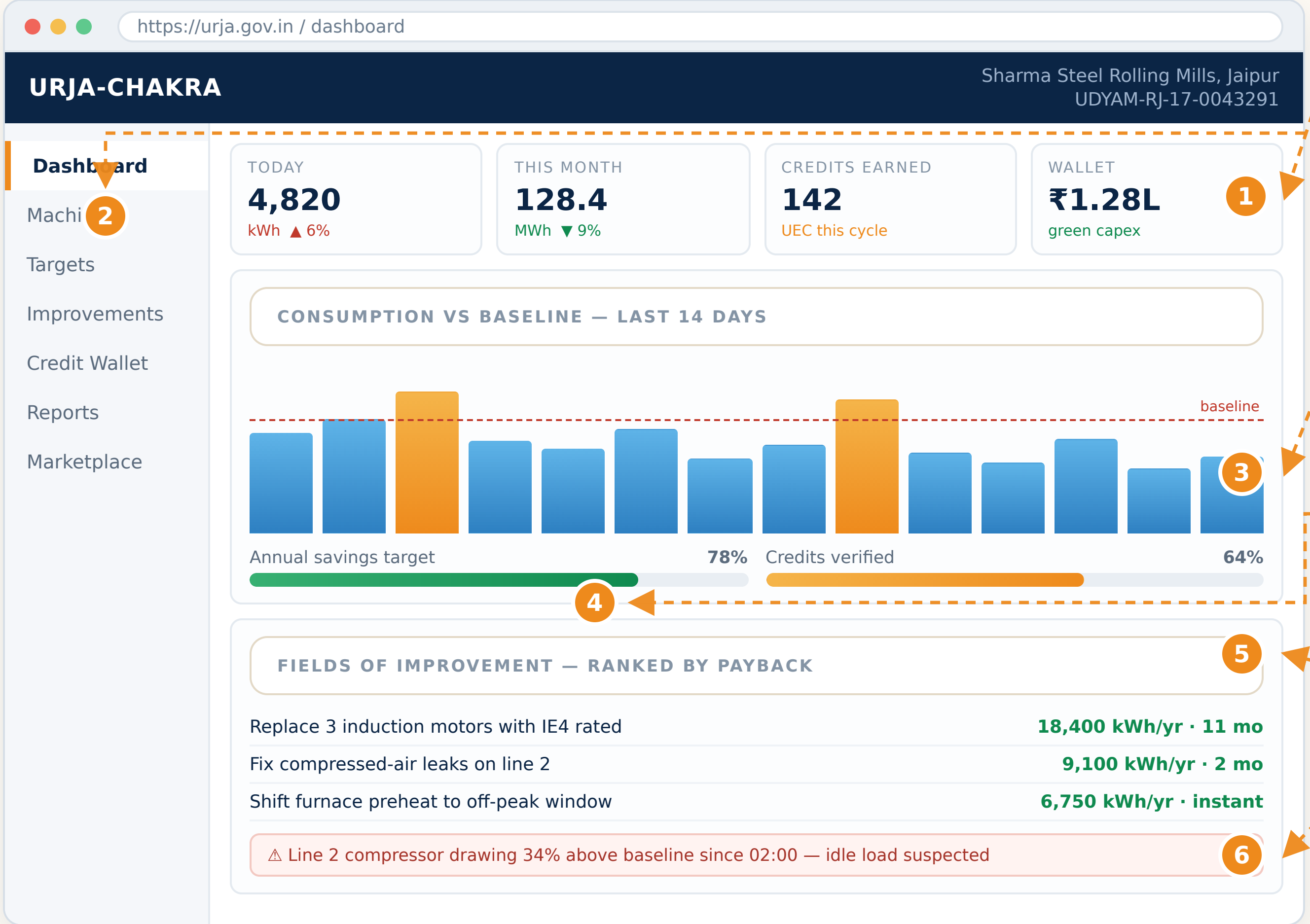
Maximum-demand penalties

One badly timed spike inflates the entire monthly tariff bill.

Equipment drift

Slowly rising current draw — wasted units and an early failure signal.

One portal, one login — what a factory owner actually sees



1. **Credit wallet**
   UEC earned, pending and tradable, with a full audit trail.
2. **Machinery registry**
   Registered assets and ratings auto-generate your savings baseline.
3. **Live consumption**
   Plant, line and machine level draw, refreshed every 15 minutes.
4. **Target tracker**
   Progress against the annual kWh savings target, updated daily.
5. **Fields of improvement**
   Ranked actions with estimated saving and payback in months.
6. **Anomaly alerts**
   Idle load, drift and tamper flagged live, not once a year.

Software skeleton — how the platform is actually programmed

FIVE-LAYER ARCHITECTURE

EDGE

Device & firmware layer

ESP32 reads the smart meter over Modbus and sub-meters via CT clamps, then publishes every 15 minutes.

MicroPython / Arduino C++ · MQTT

INGEST

Ingestion & validation layer

Broker receives readings, validates schema, signs and hash-chains each record so history cannot be rewritten.

Mosquitto · FastAPI · Pydantic

STORE

Storage layer

Time-series store for meter data; relational store for machinery registry, targets and the credit ledger.

TimescaleDB · PostgreSQL · Redis

INTEL

Intelligence layer

Baseline regression from registered machinery, LSTM forecasting, Isolation Forest for idle load and tamper detection.

pandas · scikit-learn · PyTorch

APP

Application layer

Owner dashboard, regulator console and the credit marketplace, secured by Udyam and PAN based eKYC.

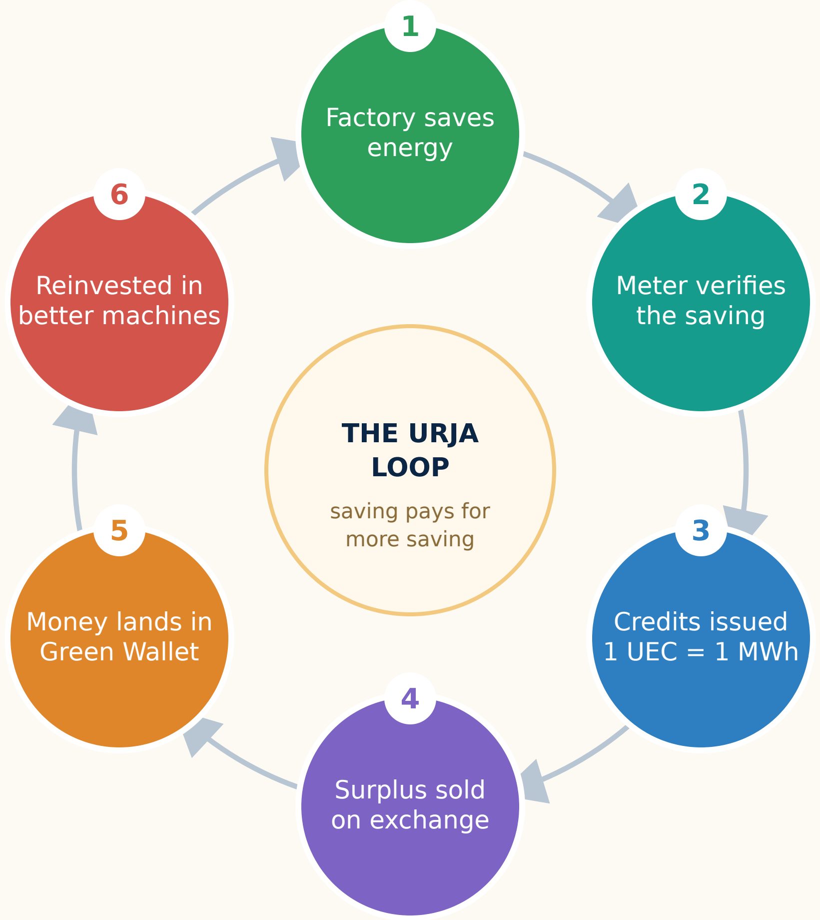
React · Grafana · JWT + OAuth2

CORE CREDIT COMPUTATION

```
# runs at the close of every monitoring period
def compute_credits(facility, period):
    kwh_actual = tsdb.sum(facility.meters, period)     # hash-chained
    output     = facility.production(period)

    # baseline is learnt from the machinery registry,
    # not a flat sector benchmark
    sec_base   = baseline_model.predict(
                     facility.machinery,               # rated loads
                     facility.shift_hours,
                     facility.history)
    kwh_base   = sec_base * output                     # expected kWh

    if anomaly.detect(facility, period).tampered:
        return audit.flag(facility, "meter_integrity")

    saved = kwh_base - kwh_actual
    if saved <= 0:
        targets.record_shortfall(facility, abs(saved))
        return

    uec = floor(saved / 1000)                       # 1 UEC = 1 MWh
    if uec < MIN_LOT:
        pool.aggregate(facility, uec)                  # MSME pooling
    else:
        registry.issue(facility, uec)

    wallet.credit(facility, uec * clearing_price,
                  restricted="energy_capex")
    return uec
```

POST /v1/telemetry	GET /v1/target	GET /v1/improvements	POST /v1/credits/trade

How an energy credit turns into money — and back into savings



WHO SELLS, WHO BUYS

Sellers	Facilities that beat their savings target, and pooled MSME blocks aggregated by the portal.
Buyers	Facilities that miss their target, plus voluntary buyers meeting green procurement or ESG commitments.
Venue	Power exchanges already used for ESCerts (IEX / PXIL), regulated by CERC.

FIXING WHAT PAT GOT WRONG

ESCerts cleared only at the floor price because supply overwhelmed demand. URJA-CHAKRA sets a **price band** — a government-backed floor that guarantees a saving is always worth something, and a ceiling so compliance never becomes ruinous.

THE GREEN WALLET — WHERE CREDIT MONEY MAY GO

- IE4 motors, variable frequency drives, LED retrofits
- Rooftop solar and waste-heat recovery capital cost
- Interest subvention on efficiency equipment loans
- Metering hardware and energy audit fees

What it changes now, and where it goes next

IMPACT

41.8%	Targets the largest block of India's electricity demand, where even a small percentage saving is an enormous absolute saving.
3 yrs to 15 minutes	Collapses the verification cycle, so a plant sees the value of an efficiency action in the same month it acts.
~1,000 to lakhs	Extends energy-credit access from a few designated consumers to the entire industrial and commercial base.
Grid	Peak-shaving across thousands of facilities defers new generation capacity and eases distribution company load.

ALIGNMENT

- **SDG 7** Clean energy
- **SDG 9** Industry
- **SDG 12** Responsible use
- **SDG 13** Climate action
- **EC Act** 2001 & 2022

FUTURE EXPANSION — BEYOND THE FACTORY FLOOR

PHASE 1 — Manufacturing & MSME clusters
Rolling mills, textile units, foundries and industrial estates — the highest energy intensity per site.

PHASE 2 — Commercial buildings
Malls, office towers, IT parks and hotels, where HVAC alone dominates the load profile.

PHASE 3 — Institutional campuses
Universities, hostels, hospitals and government buildings — with wing-level leaderboards to drive behaviour change.

PHASE 4 — Cold chain, warehousing & data centres
Continuous-load facilities where efficiency gains compound fastest, and the fastest growing demand segment.

PHASE 5 — Housing societies & agriculture
Residential blocks and irrigation pump sets, using the same meter, target and credit rails.

LATER — Interoperability
Credits made convertible into CCTS carbon credits once savings are mapped to grid emission factors — energy first, carbon later.

Sources

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